k-space diagonalisation, Zak phase and cavity dressing of the SSH chain

Step-by-step computational recipe behind `k_space_diag.ipynb`. Every step states its **input**, the **formulas**, the **output**, and a **check** that pins it down numerically. Written so the whole thing can be re-implemented from the maths alone.

System: the `PBC=True` bosonic `PHOTONICChain` of `packages/qcg-quantum-design/src/ssh_chain_sq.py` — an SSH chain of N unit cells (two transmons each) with capacitive and cavity-mediated Josephson coupling.

Part 0 — Model, indices and conventions

0.1 Indices

- N unit cells, $n = 2N$ transmon modes.
- Site index $l = 2j + \sigma$, with $j = 0 \dots N-1$ the cell and $\sigma \in \{0, 1\} = \{A, B\}$ the sublattice.
- Nambu vector $\Phi = (b_0, \dots, b_{n-1},\ b_0^\dagger, \dots, b_{n-1}^\dagger)^T$, length $2n$.

0.2 Coupling constants

$$g_{c1} = \tfrac{1}{4}\Omega_C(1+\chi_C), \qquad g_{c2} = \tfrac{1}{4}\Omega_C(1-\chi_C),$$

$$g_{J1} = \Omega_J(1+\chi_J), \qquad g_{J2} = -\Omega_J(1-\chi_J), \qquad g_J \equiv g_{J1} + g_{J2}.$$

Note $g_{c1} + g_{c2} = \tfrac{1}{2}\Omega_C$ always.

0.3 The two normalisation conventions (this trips everything up — fix it once)

Honest / Nambu convention. $H = \tfrac{1}{2}\Phi^\dagger \mathcal{H} \Phi + \text{const}$. Here A has diagonal $\tfrac{1}{2}$ and the bare transmon frequency reads $\tfrac{1}{2}$.

Physical convention. Multiply every block by 2 (this is `_nambu_prefactor = 2` in `ssh_chain_sq`). The bare transmon then sits at frequency 1. All *reported* frequencies, all Hamiltonians fed to the cavity problem, and the analytic block formulas of §1.5 use this one.

Rule of thumb: **build** in the honest convention (so the $\tfrac{1}{2}\Phi^\dagger\mathcal{H}\Phi$ algebra and τ_3 diagonalisation are standard), **report** in the physical one.

Part I — Momentum space and block diagonalisation by permutation

Step 1 — Real-space Nambu blocks

Input: $N, \Omega_C, \chi_C, \Omega_J, \chi_J$.

Build the SSH hopping matrix T ($n \times n$, real symmetric):

$$T_{l,l+1} = T_{l+1,l} = \begin{cases} g_{c1} & l \text{ even (intra-cell)} \\ g_{c2} & l \text{ odd (inter-cell)} \end{cases} \qquad l = 0 \dots n-2, \qquad T_{0,n-1} = T_{n-1,0} = g_{c2} \ \text{(PBC wrap)}.$$

Then

$$A = \tfrac{1}{2}\big(I + T\big), \qquad B = -\tfrac{1}{2}T.$$

Identically for Josephson, with T_J built the same way from g_{J1}, g_{J2}:

$$A_J = \tfrac{1}{2}\big(g_J I - T_J\big), \qquad B_J = -\tfrac{1}{2}T_J - \tfrac{1}{2}g_J I.$$

Assemble the $2n \times 2n$ Nambu matrices

$$\mathcal{H}_C = \begin{pmatrix} A & B \\ B & A \end{pmatrix}, \qquad \mathcal{H}_J = \begin{pmatrix} A_J & B_J \\ B_J & A_J \end{pmatrix}.$$

Output: $\mathcal{H}_C, \mathcal{H}_J$, both $2n \times 2n$ real symmetric.

Check: these are exactly what `PHOTONICChain.__init__` builds (`D_herm`, `H_J_herm`).

Step 2 — The Fourier transform as a change of operator basis

Input: N.

Momenta on the FFT grid (this matters — it is the grid on which $-k$ lands on another grid point):

$$k_m = 2\pi\,\texttt{fftfreq}(N)_m \quad \longrightarrow \quad \texttt{fftshift} \text{ onto } (-\pi, \pi].$$

The transform itself is

$$b_{j\sigma} = \frac{1}{\sqrt{N}} \sum_m e^{i k_m j}\, c_{m\sigma}.$$

As a matrix, define $W_{jm} = e^{i k_m j}/\sqrt{N}$ ($N \times N$) and

$$\mathcal{F}_{(2j+\sigma),\,(2m+\sigma')} = W_{jm}\,\delta_{\sigma\sigma'} \qquad (n \times n,\ \text{unitary}),$$

i.e. $\mathcal{F}$ is W replicated on the two sublattice diagonal blocks. Then $b = \mathcal{F}c$ and $b^\dagger = \mathcal{F}^* c^\dagger$, so the Nambu lift is

$$\mathcal{G} = \begin{pmatrix} \mathcal{F} & 0 \\ 0 & \mathcal{F}^* \end{pmatrix}, \qquad \boxed{\mathcal{H}^k = \mathcal{G}^\dagger\,\mathcal{H}\,\mathcal{G}}$$

Output: $\mathcal{H}^k_C, \mathcal{H}^k_J$, both $2n \times 2n$.

Check: $\mathcal{F}^\dagger\mathcal{F} = I$; the spectrum of $\mathcal{H}^k$ equals that of $\mathcal{H}$ to $\sim 10^{-16}$.

Step 3 — The permutation that block-diagonalises

Input: the momentum list $\{k_m\}$.

For each m, find m' with $k_{m'} \equiv -k_m \pmod{2\pi}$. Then walk $m = 0 \dots N-1$, skipping already-used indices, and emit an index group:

$$\text{TRIM } (m' = m): \quad \big[\,2m,\ 2m{+}1,\ n{+}2m,\ n{+}2m{+}1\,\big] \quad (4 \text{ indices})$$

$$\text{generic}: \quad \big[\,2m,\ 2m{+}1,\ 2m',\ 2m'{+}1,\ n{+}2m,\ n{+}2m{+}1,\ n{+}2m',\ n{+}2m'{+}1\,\big] \qquad (8)$$

Concatenate the groups into a permutation P, and apply it symmetrically: $\mathcal{H}^{\mathrm{bd}} = \mathcal{H}^k[P][:, P]$.

Why it works. $\mathcal{F}^\dagger A \mathcal{F}$ is diagonal in k (translation invariance); $\mathcal{F}^\dagger B \mathcal{F}^*$ connects $k \leftrightarrow -k$ only (a zero-momentum pair). So the sparsity graph of $\mathcal{H}^k$ splits into connected components labelled by the unordered pair $\{k, -k\}$ — that is exactly what a permutation can expose, and nothing more.

Output: block-diagonal $\mathcal{H}^{\mathrm{bd}}$: 4×4 blocks at TRIMs ($k = 0$, and $k = \pi$ if N even), 8×8 elsewhere. Block sizes sum to $2n$.

Check: Frobenius norm of everything outside the declared blocks $\sim 10^{-15}$.

Step 4 — The per-k blocks in closed form

Define the Bloch factors

$$\delta_{C,k} = g_{c1} + g_{c2} e^{-ik}, \qquad \delta_{J,k} = g_{J1} + g_{J2} e^{-ik},$$

and note $\delta_{C,-k} = \delta_{C,k}^*$ and $|\delta_{C,k}|^2 = g_{c1}^2 + g_{c2}^2 + 2 g_{c1} g_{c2} \cos k$.

Carrying the sums $\sum_d \alpha_{\sigma\sigma'}(d) e^{-ikd}$ through gives, in the honest convention,

$$\tilde{A}(k) = \tfrac{1}{2} \begin{pmatrix} 1 & \delta_{C,k} \\ \delta_{C,k}^* & 1 \end{pmatrix}, \qquad \tilde{B}(k) = -\tfrac{1}{2} \begin{pmatrix} 0 & \delta_{C,k} \\ \delta_{C,k}^* & 0 \end{pmatrix},$$

$$\tilde{A}_J(k) = \tfrac{1}{2} \begin{pmatrix} g_J & -\delta_{J,k} \\ -\delta_{J,k}^* & g_J \end{pmatrix}, \qquad \tilde{B}_J(k) = -\tfrac{1}{2} \begin{pmatrix} g_J & \delta_{J,k} \\ \delta_{J,k}^* & g_J \end{pmatrix}.$$

The generic-k 8×8 sector, ordered $(c_{kA}, c_{kB}, c_{-kA}, c_{-kB})$ then daggers, is

$$\mathcal{H}^{(k)} = \begin{pmatrix} \mathcal{A} & \mathcal{B} \\ \mathcal{B}^* & \mathcal{A}^* \end{pmatrix}, \quad \mathcal{A} = \begin{pmatrix} \tilde{A}(k) & 0 \\ 0 & \tilde{A}(-k) \end{pmatrix}, \quad \mathcal{B} = \begin{pmatrix} 0 & \tilde{B}(k) \\ \tilde{B}(-k) & 0 \end{pmatrix}.$$

All-annihilation / factor-2 form (what `k_space_hamiltonians` returns; multiply by 2 and drop the redundant creation half). With $\bar{\delta} \equiv \delta^*$:

$$A_C = \begin{pmatrix} 1 & \delta_C & & \\ \bar{\delta}_C & 1 & & \\ & & 1 & \bar{\delta}_C \\ & & \delta_C & 1 \end{pmatrix}, \qquad B_C = \begin{pmatrix} & & & -\delta_C \\ & & -\bar{\delta}_C & \\ & -\bar{\delta}_C & & \\ -\delta_C & & & \end{pmatrix},$$

$$A_J = \begin{pmatrix} g_J & -\delta_J & & \\ -\bar{\delta}_J & g_J & & \\ & & g_J & -\bar{\delta}_J \\ & & -\delta_J & g_J \end{pmatrix}, \qquad B_J = \begin{pmatrix} & & -g_J & -\delta_J \\ & & -\bar{\delta}_J & -g_J \\ -g_J & -\bar{\delta}_J & & \\ -\delta_J & -g_J & & \end{pmatrix}.$$

At a TRIM ($\sin k = 0$, so δ real) the sector collapses to 2×2:

$$A_C = \begin{pmatrix} 1 & \delta_C \\ \bar{\delta}_C & 1 \end{pmatrix}, \; B_C = -\begin{pmatrix} 0 & \delta_C \\ \bar{\delta}_C & 0 \end{pmatrix}, \; A_J = \begin{pmatrix} g_J & -\delta_J \\ -\bar{\delta}_J & g_J \end{pmatrix}, \; B_J = -\begin{pmatrix} g_J & \delta_J \\ \bar{\delta}_J & g_J \end{pmatrix}.$$

Three traps, all hit in the first hand-derivation:

1. the TRIM B_C carries $-\delta_C$, the same sign as the generic case;
2. the generic B_C B-sublattice entries are $-\bar{\delta}_C$, **not** $-\delta_C$ — the anomalous coefficient of $c^\dagger_{kB}c^\dagger_{-kA}$ is $\tilde{B}_{BA}(k)+\tilde{B}_{AB}(-k)=-\delta^*_C$;
3. the $-g_J$ on-site Josephson pairing lives on the $k\leftrightarrow -k$ block-diagonal (entries $[0,2],[1,3]$ and transposes), **not** on the main diagonal — on-site pairing $b^\dagger_l b^\dagger_l$ Fourier-transforms to $c^\dagger_{k\sigma}c^\dagger_{-k\sigma}$. It only *looks* diagonal at a TRIM because there $k=-k$.

Also: the TRIM test must be $\sin k=0$, not `k in {0, pi}` — with `fftshift` the second TRIM sits at $k=-\pi$.

Check: these agree block-by-block with $\mathcal{G}^\dagger\mathcal{H}\mathcal{G}$ to $\lesssim 10^{-15}$.

Part II — Diagonalising H_C sector by sector

Everything here is a **canonical (paraunitary) transform**: a matrix $\mathcal{T}$ with

$$\mathcal{T}^\dagger\tau_3\mathcal{T}=\tau_3,\qquad \tau_3=\mathrm{diag}(I_m,-I_m),$$

acting as $\mathcal{H}\to\mathcal{T}^\dagger\mathcal{H}\,\mathcal{T}$. ($m$ = 2 at a TRIM, 4 generic.)

Step 5 — Phased sublattice rotation

Input: k, g_{c1}, g_{c2}. Set $\phi_k=\arg\delta_{C,k}$.

The naive symmetric/antisymmetric combination $(b_{kA}\pm b_{kB})/\sqrt{2}$ **fails** for generic k: it leaves an α_k–β_k number-conserving coupling of magnitude $\frac{1}{2}g_{c2}|\sin k|$ (zero only where $\delta_{C,k}$ is real, i.e. at the TRIMs). The correct combination is the eigenbasis of $\tilde{A}(k)$:

$$\boxed{\alpha_k=\tfrac{1}{\sqrt{2}}\big(e^{-i\phi_k/2}b_{kA}+e^{+i\phi_k/2}b_{kB}\big),\qquad \beta_k=\tfrac{1}{\sqrt{2}}\big(e^{-i\phi_k/2}b_{kA}-e^{+i\phi_k/2}b_{kB}\big)}$$

so that $\tilde{A}(k)\to\mathrm{diag}\big(\frac{1}{2}(1+|\delta_{C,k}|),\ \frac{1}{2}(1-|\delta_{C,k}|)\big)$.

As a matrix, rows $=(\alpha,\beta)$:

$$R(k)=\frac{1}{\sqrt{2}}\begin{pmatrix}e^{-i\phi_k/2} & e^{+i\phi_k/2}\\ e^{-i\phi_k/2} & -e^{+i\phi_k/2}\end{pmatrix},\qquad R(-k)=R(k)\big|_{\phi_k\to-\phi_k},$$

and for a generic sector $M=\mathrm{blkdiag}\,\big(R(k),R(-k)\big)$ (4×4).

Nambu lift of a passive rotation. If the new modes are $\mu=Mb$ then $b=M^{-1}\mu$, so

$$\mathcal{T}_{\rm rot}=\begin{pmatrix}M^{-1} & 0\\ 0 & (M^{-1})^*\end{pmatrix}.$$

(Getting this dagger backwards silently produces a wrong, non-diagonalising rotation — worth an explicit test that the number-off-diagonal really drops to $\sim 10^{-16}$.)

Step 6 — Two-mode squeeze

Input: the rotated sector.

After Step 5 each $\{k, -k\}$ pair is, per species $\gamma \in \{\alpha, \beta\}$,

$$\omega_\gamma\big(n_{k\gamma} + n_{-k\gamma}\big) + \lambda_\gamma\big(\gamma_k^\dagger \gamma_{-k}^\dagger + \text{h.c.}\big),$$

$$\omega_\alpha = \tfrac{1}{2}(1 + |\delta_{C,k}|),\ \ \lambda_\alpha = -\tfrac{1}{2}|\delta_{C,k}|, \qquad \omega_\beta = \tfrac{1}{2}(1 - |\delta_{C,k}|),\ \ \lambda_\beta = +\tfrac{1}{2}|\delta_{C,k}|.$$

A two-mode squeeze $S\gamma_i S^\dagger = \cosh z\, \gamma_i + \sinh z\, \gamma_j^\dagger$ has Nambu form

$$\mathcal{T}_{\text{sq}} = \begin{pmatrix} U & V \\ V^* & U^* \end{pmatrix}, \quad U_{ii} = U_{jj} = \cosh z,\ \ V_{ij} = V_{ji} = \sinh z,\ \ \text{rest } U = I,\ V = 0.$$

The generic 2×2 problem $\left(\begin{smallmatrix} \omega & \lambda \\ \lambda & \omega \end{smallmatrix}\right)$ is killed by $z = \frac{1}{4} \ln \frac{\omega+\lambda}{\omega-\lambda}$, leaving $\sqrt{\omega^2 - \lambda^2}$. Substituting $\omega_\pm, \lambda_\pm$:

$$\boxed{z_{k,\alpha} = \tfrac{1}{4} \ln\big(1 + 2|\delta_{C,k}|\big), \qquad z_{k,\beta} = \tfrac{1}{4} \ln\big(1 - 2|\delta_{C,k}|\big)}$$

applied on the pairs (α_k, α_{-k}) and (β_k, β_{-k}). At a TRIM $k = -k$ and this degenerates into a **single-mode** squeeze ($i = j$), same z.

The **factor 2 inside the log is essential**: z_β requires $1 - 2|\delta_{C,k}| > 0$, and since $\max_k |\delta_{C,k}| = g_{c1} + g_{c2}$ that is exactly the stated existence bound

$$g_{c1} + g_{c2} < \tfrac{1}{2}.$$

(If one instead writes δ with the factor 2 absorbed, the same formulas read $z = \frac{1}{4} \ln(1 \pm |\delta|)$ — the bound is unchanged, only the definition of δ moved.)

Total transform: $\mathcal{T} = \mathcal{T}_{\text{rot}}\, \mathcal{T}_{\text{sq}}$, and $\mathcal{T}^\dagger \mathcal{H}_C^{(k)} \mathcal{T}$ is diagonal.

Bands (physical convention, $\times 2$):

$$\boxed{\omega_{k,\alpha} = \sqrt{1 + 2|\delta_{C,k}|}, \qquad \omega_{k,\beta} = \sqrt{1 - 2|\delta_{C,k}|}}$$

Checks (all three must agree to $\sim 10^{-15}$):

1. $2\times$ the diagonal of $\mathcal{T}^\dagger \mathcal{H}_C^{(k)} \mathcal{T}$;
2. the positive eigenvalues of $\tau_3\, \mathcal{H}_C^{(k)}$ — the convention-independent definition of bosonic BdG frequencies, times the same factor 2;
3. `PHOTONICChain(PBC=True)._nambu_prefactor() * diag(chain.H_C)`.

Also check paraunitarity: $\|\mathcal{T}^\dagger \tau_3 \mathcal{T} - \tau_3\| < 10^{-9}$.

The same $\mathcal{T}$ does not diagonalise H_J — $\arg \delta_{J,k} \neq \phi_k$ — so

$$\tilde{\mathcal{H}}_J(k) = \mathcal{T}^\dagger \mathcal{H}_J^{(k)} \mathcal{T} = \begin{pmatrix} \tilde{\mathcal{A}}_J & \tilde{\mathcal{B}}_J \\ \tilde{\mathcal{B}}_J^* & \tilde{\mathcal{A}}_J^* \end{pmatrix}$$

retains an α–β number mixing $\tilde{\mathcal{A}}_J$ *and* a pairing block $\tilde{\mathcal{B}}_J$. These two objects drive Parts IV–V.

Part III — Zak phase

Step 7 — Discrete Berry phase (Wilson loop) with the orbital gauge

Formula. Use the single-log-of-the-product form — it is gauge invariant (each $|u_k\rangle$ appears once as bra and once as ket, so the arbitrary phases `eigh` assigns cancel) and folds the answer into $(-\pi, \pi]$ automatically:

$$\boxed{\gamma = -\operatorname{Im}\ln\Big[\textstyle\prod_j \langle u(k_j)\,|\,u(k_{j+1})\rangle\Big]}$$

(Writing it as $-\prod_j \operatorname{Im}\ln\langle\cdot\rangle$ is wrong; the sum form $-\sum_j \operatorname{Im}\ln$ is equivalent modulo 2π but exposed to branch cuts.)

Grid. $k_j = 2\pi j/N_k$, $j = 0 \ldots N_k - 1$, endpoint excluded — $k = 0$ and $k = 2\pi$ are the same point; the wrap is done by hand in the closure term.

Inner product. $\langle a|b\rangle = \sum_i \bar{a}_i b_i$ (conjugate the **bra**).

Orbital gauge. The loop is over the cell-periodic part $|u_k\rangle = e^{-ik\hat{r}}|\psi_k\rangle$ with $\hat{r} = \operatorname{diag}(s_A, s_B)$ the sublattice positions in the cell. Since $u_{k+2\pi} = e^{-i2\pi\hat{r}}u_k$, the entire orbital winding can be lumped into the closing link:

$$\gamma = -\operatorname{Im}\ln\Big[\prod_{j=0}^{N_k-2}\langle\psi_j|\psi_{j+1}\rangle \;\cdot\; \langle\psi_{N_k-1}|\,\hat{g}\,|\psi_0\rangle\Big], \qquad \hat{g} = \operatorname{diag}\left(e^{-2\pi i s_A},\, e^{-2\pi i s_B}\right).$$

Lumping is exact whenever the sublattice weights $|\psi_{k,\sigma}|^2$ are k-independent (true for a chiral SSH band); otherwise distribute $e^{-i\Delta k\,\hat{r}}$ over every link instead (difference is $O(1/N_k)$).

States. $|\psi_k\rangle$ = eigenvector of the SSH Bloch Hamiltonian

$$h(k) = \begin{pmatrix} 0 & \delta_{C,k} \\ \delta_{C,k}^* & 0 \end{pmatrix} \;=\; 2\tilde{A}(k) - I,$$

lower band = eigenvalue $-|\delta_{C,k}|$, upper band = $+|\delta_{C,k}|$. (Same eigenvectors as $\tilde{A}(k)$; the squeeze never rotates sublattice space, so these are also the quasiparticle sublattice spinors.)

Result. For the chiral SSH band $|\psi_{k,A}|^2 = |\psi_{k,B}|^2 = \frac{1}{2}$, so the orbital term contributes $-\pi(s_A + s_B)$:

$s = (s_A, s_B)$	γ **at** $\chi_C > 0$	γ **at** $\chi_C < 0$
$(0, 0)$ periodic gauge	0	π
$(-\frac{1}{4}, \frac{1}{4})$ symmetric cell	0	π
$(0, \frac{1}{2})$	$\pi/2$	$-\pi/2$

Exactly quantised and grid-independent ($N_k = 10, 13, 50, 400$ identical); transition at $\chi_C = 0$, where $g_{c1} = g_{c2}$ and the gap closes at $k = \pi$.

Step 8 — Does the pairing change it? (BdG Wilson loop)

Question: the squeeze mixes b and $b^\dagger$; does the Zak phase of the *quasiparticle* band differ from that of the bare SSH band?

Setup. Single-k BdG Bloch Hamiltonian on $\Psi_k = (b_{kA}, b_{kB}, b^\dagger_{-kA}, b^\dagger_{-kB})$, physical convention:

$$\mathcal{H}_{\mathrm{BdG}}(k) = \begin{pmatrix} h(k) & \Delta(k) \\ \Delta(k) & h(k) \end{pmatrix}, \qquad h(k) = \begin{pmatrix} 1 & \delta_{C,k} \\ \delta_{C,k}^* & 1 \end{pmatrix}, \qquad \Delta(k) = I - h(k).$$

Diagonalise the **non-Hermitian** $\tau_3\mathcal{H}_{\mathrm{BdG}}(k)$ ($\tau_3 = \mathrm{diag}(1,1,-1,-1)$), pick the eigenvalue nearest $+\sqrt{1-2|\delta_{C,k}|}$ (the β band), and normalise with the **para-norm**

$$\Phi^\dagger \tau_3 \Phi = +1 \qquad (\text{i.e. } |u|^2 - |v|^2 = 1),$$

not the ordinary norm. Run the same Wilson loop on the 4-component $\Phi_\beta(k)$, with the hole block taking the *opposite* orbital phase:

$$\hat{g} = \mathrm{diag}\left(e^{-2\pi i s_A},\, e^{-2\pi i s_B},\, e^{+2\pi i s_A},\, e^{+2\pi i s_B}\right),$$

because v_σ transforms like $b^\dagger_{-k,\sigma}$. (Magnitudes drop out of $\mathrm{Im}\ln$, so the non-unit ordinary norm is harmless.)

Answer: unchanged, to $\sim 10^{-12}$, in every orbital gauge and on both sides of the transition. Reason: $\Delta(k) = I - h(k)$ **commutes with** $h(k)$, so the Bogoliubov rotation is (phased sublattice rotation) $\times$ (real squeeze on the $\{k,-k\}$ pair). The particle block (u_{kA}, u_{kB}) is exactly the SSH eigenvector, the hole block points the same way with a real, non-winding amplitude $\sinh z_{k,\beta}$, and chiral symmetry survives ($|u_{kA}|^2 = |u_{kB}|^2$ and $|v_{kA}|^2 = |v_{kB}|^2$ at every k). Nothing in the squeeze can wind.

Part IV — Schrieffer–Wolff correction

Step 9 — What is actually eliminated

Add the **cavity resonator** (the mode mediating the inductive coupling — not a readout resonator):

$$H = \omega_r a^\dagger a + H_C - \gamma\,(a + a^\dagger)\,H_J.$$

`PHOTONICChain.schrieffer_wolff_correction` **does not eliminate the cavity.** It eliminates only the *matter-non-number-conserving* part of the coupling — the pairing blocks $\tilde{\mathcal{B}}_J$, which would take the single quasiparticle into the 3-QP sector — and folds it into a 2nd-order term living in the **same composite space** as H,

$$(\text{cavity Fock } d_{\mathrm{ph}}) \;\otimes\; (\text{single-QP matter } d_X),$$

added as `H = H + H_SW`. The number-conserving $\tilde{\mathcal{A}}_J$ stays in H at first order.

Ingredients, per $\{k,-k\}$ sector (physical convention, m = 4 generic / 2 TRIM QP modes):

- QP frequencies $w_i = 2\,[\mathcal{T}^\dagger \mathcal{H}_C^{(k)} \mathcal{T}]_{ii}$, $i = 1 \ldots m$, ordered $(\alpha_k, \beta_k, \alpha_{-k}, \beta_{-k})$;
- pairing block $\mathcal{B} \equiv 2\,[\tilde{\mathcal{H}}_J(k)]_{[:m,\,m:]}$, giving the operator $\hat{B} = \sum_{ij} \mathcal{B}_{ij}\,\gamma_i^\dagger \gamma_j^\dagger$;
- single-QP basis $|X\rangle = \gamma_X^\dagger |0\rangle$ with $E_X = w_X$ ($d_X = m$ states);
- virtual basis: **3-QP** occupation states $|r\rangle$ with $E_r = \sum_i n_i^{(r)} w_i$ (the -2 branch would need -1 excitations and is skipped);
- matrix elements $V_{rX} = \langle r|\hat{B}|X\rangle$, evaluated in the bosonic Fock basis;
- photon ladder $s = \gamma(a + a^\dagger)$, i.e. $s_{n,n'} = \gamma\left(\sqrt{n'}\,\delta_{n,n'-1} + \sqrt{n'+1}\,\delta_{n,n'+1}\right)$.

The formula. With composite index $(n, X) \to n\,d_X + X$ (photon outer, matter inner):

$$H_{\mathrm{SW}}\big[(n_1,X),(n_2,X')\big] = \sum_{n_{\mathrm{mid}}} s_{n_1 n_{\mathrm{mid}}}\, s_{n_{\mathrm{mid}} n_2}\, \Big(\mathrm{T1}_{XX'} - \mathrm{T2}_{XX'}\Big),$$

$$\mathrm{T1}_{XX'} = \sum_r \frac{V_{rX}\,\overline{V_{rX'}}}{(n_1 - n_{\mathrm{mid}})\,\omega_r + E_X - E_r}, \qquad \mathrm{T2}_{XX'} = \sum_r \frac{V_{rX}\,\overline{V_{rX'}}}{(n_{\mathrm{mid}} - n_2)\,\omega_r + E_r - E_{X'}},$$

then symmetrise and halve:

$$H_{\mathrm{SW}} \;\leftarrow\; \tfrac{1}{2}\big(H_{\mathrm{SW}} + H_{\mathrm{SW}}^\dagger\big), \qquad H_{\mathrm{SW}} \;\leftarrow\; \tfrac{1}{2}\,H_{\mathrm{SW}}.$$

Guard denominators with $|\cdot| > \varepsilon$ (drop the term otherwise). Because $s_{nn'}$ is purely off-diagonal in photon number, $s_{n_1 n_{\mathrm{mid}}}\, s_{n_{\mathrm{mid}} n_2} \neq 0$ only for $|n_1 - n_2| \in \{0, 2\}$ — the correction has a dispersive ($\Delta n_{\mathrm{ph}} = 0$) piece and a cavity two-photon squeezing ($\Delta n_{\mathrm{ph}} = \pm 2$, $\propto a^{\dagger 2}, a^2$) piece.

Exact γ^2 scaling: γ enters only through s, quadratically. So H_{SW}/γ^2 is γ-independent by construction — build it once at $\gamma = 1$ and rescale.

Step 9b — The split, and why the block structure is simpler than expected

> **Observation (read off directly from the composite matrix plots).** H_{SW} has **no $+k \to -k$ matrix element at all** — those entries are identically zero, to machine precision, at every k and for every parameter set tried. Its off-diagonal part is *purely* $\alpha_k \leftrightarrow \beta_k$ **at the same momentum**.
>
> So "k-changing" is a misnomer for it: the correction never changes k, it only flips the band. A better label is **interband**.

The immediate consequence is that the block structure of H_{SW} is **finer than the $\{k, -k\}$ sector** the rest of the calculation is organised around. Inside one 8×8 sector, H_{SW} does not connect the $+k$ half to the $-k$ half at all, so it further decomposes into two *independent* blocks, one per momentum:

$$H_{\mathrm{SW}}^{(\{k,-k\})} = H_{\mathrm{SW}}^{(+k)} \;\oplus\; H_{\mathrm{SW}}^{(-k)}, \qquad H_{\mathrm{SW}}^{(\pm k)} \text{ acting on } \{\alpha_{\pm k}, \beta_{\pm k}\} \otimes (\text{cavity Fock}),$$

each of dimension $2d_{\mathrm{ph}}$. Nothing couples the two. (Sanity check to reproduce:
$\|H_{\mathrm{SW}}[\,\{\alpha_k, \beta_k\},\ \{\alpha_{-k}, \beta_{-k}\}\,]\| = 0$ exactly — not small, zero.)

Within one such block the split is then simply diagonal vs off-diagonal in the band index:

	matrix elements	**share of $\lvert H_{\mathrm{SW}}\rvert$**
band-diagonal ("k-preserving")	$\gamma_{k\gamma}^\dagger \gamma_{k\gamma} \otimes (\cdot)_{\mathrm{cav}}$	$\approx 99\%$
band-off-diagonal ("k-changing", really *interband*)	$\gamma_{k\alpha}^\dagger \gamma_{k\beta} \otimes (\cdot)_{\mathrm{cav}}$	10^{-3}–10^{-2}

The interband part is small because $\tilde{\mathcal{B}}_J$ predominantly pairs α_k with β_{-k}; only its same-band entries feed a band flip.

Why does the $\pm k$ mixing vanish? A quick argument: the cavity carries no momentum and $\hat{B}, \hat{B}^\dagger$ create/destroy zero-momentum pairs, so total crystal momentum is conserved and a single QP starting at $+k$ must return to $+k$. Whether that is the whole story — in particular why the cancellation is *exact* rather than merely small, and whether it survives away from this coupling structure — is left open here; see the open points at the end.

Step 9c — Interband resonance condition

The interband gap and the resonant momentum, solved in closed form. From $\sqrt{1+2x}-\sqrt{1-2x}=\omega_r$ (square twice):

$$x \equiv |\delta_{C,k^\star}| = \frac{\omega_r}{2}\sqrt{1-\frac{\omega_r^2}{4}}, \qquad \cos k^\star = \frac{x^2-g_{c1}^2-g_{c2}^2}{2g_{c1}g_{c2}}.$$

The gap sweeps $\big[\,\mathrm{gap}(\pi),\ \mathrm{gap}(0)\,\big]$ across the BZ with $\mathrm{gap}(0)=\sqrt{1+2(g_{c1}+g_{c2})}-\sqrt{1-2(g_{c1}+g_{c2})}\approx 2(g_{c1}+g_{c2})=\Omega_C$. Choosing $\omega_r=2(g_{c1}+g_{c2})$ puts $k^\star$ just inside the zone (for the reference parameters, $k^\star=0.2315=0.0737\pi$).

Step 9d — Sweep of the interband ("k-changing") part

For each (χ_C,Ω_C) with $\omega_r=\Omega_C$, sweep k over the BZ and record $\max_k \big|H_{\mathrm{SW}}^{k\text{-chg}}(k)\big|/\gamma^2$ (log colour scale). Findings:

- it **vanishes identically on** $\chi_C=0$ ($\lesssim 10^{-14}$) — there $g_{c1}=g_{c2}$, $\tilde{\mathcal{B}}_J$ is exactly band-diagonal, and $\alpha_k\to\beta_k$ is forbidden;
- dimerisation switches it on; grows monotonically with $|\chi_C|$ and Ω_C, symmetric under $\chi_C\to-\chi_C$;
- peak ≈ 3.0 (in γ^{-2} units) at $(\chi_C,\Omega_C)=(-0.70,0.64)$.

Part V — Cavity-dressed bands and the topological second band

Step 10 — The effective Hamiltonian per k

Restrict to the $+k$ quasiparticles. By Step 9b this is an **exact block decomposition, not a truncation**: neither the direct coupling nor H_{SW} has a single matrix element linking $+k$ to $-k$, so the $-k$ half simply factors out. Work on

$$\{\alpha_k,\beta_k\}\otimes(\text{cavity Fock } d_{\mathrm{ph}}), \qquad \text{index } = n\cdot 2+s,\ \ s\in\{0=\alpha,\,1=\beta\},$$

a $2d_{\mathrm{ph}}$-dimensional space. Then

$$\boxed{H_{\mathrm{eff}}(k)=\underbrace{\omega_r\,a^\dagger a+\mathrm{diag}(\omega_{k\alpha},\omega_{k\beta})-\gamma\,(a+a^\dagger)\otimes M}_{\texttt{build_full_hamiltonian},\ \text{single-exc subspace}}\ +\ H_{\mathrm{SW}}^{+k}(k)}$$

with

- $M=\tilde{\mathcal{A}}_J+\tilde{\mathcal{A}}_J^\dagger$ restricted to the $+k$ band block — this reproduces `build_HJ_fock`'s coefficient $A_J+D_J^T$, since the lower Nambu block satisfies $D_J=\mathcal{A}_J^*$;
- $(a+a^\dagger)_{nn'}=\sqrt{n'}\,\delta_{n,n'-1}+\sqrt{n'+1}\,\delta_{n,n'+1}$;
- H_{SW}^{+k} = Step 9's H_{SW} restricted to rows/cols whose QP mode index is α_k or β_k — **all of it**, band-diagonal *and* interband, since by Step 9b nothing in H_{SW} leaves the $+k$ block.

H_{eff} is Hermitian; diagonalise it normally.

What opens the gap: the direct term $-\gamma(a+a^\dagger)\otimes M$, whose interband element drives a Jaynes–Cummings anticrossing between $|\alpha_k,0\rangle$ and $|\beta_k,1\rangle$ exactly at $k^\star$. Gap size $\approx 2\gamma\,|M_{\alpha\beta}(k^\star)|$.

Step 11 — Composite inner product and Zak phase

An eigenstate is

$$|\psi_n(k)\rangle = |\phi_\alpha^{(n)}(k)\rangle_{\text{cav}} \otimes |\alpha_k\rangle + |\phi_\beta^{(n)}(k)\rangle_{\text{cav}} \otimes |\beta_k\rangle.$$

The band kets are k-dependent, so overlaps at different k do **not** reduce to the eigenvector components alone:

$$\langle\psi_n(k)|\psi_n(k')\rangle = \sum_{s,s'} \underbrace{\langle u_s(k)|u_{s'}(k')\rangle_{\text{sub}}}_{2\times 2}\ \underbrace{\langle\phi_s^{(n)}(k)|\phi_{s'}^{(n)}(k')\rangle_{\text{cav}}}_{\text{Fock, } k\text{-indep}}.$$

Practical recipe — lift into a fixed basis. Reshape the eigenvector to $C^{(n)}(k) \in \mathbb{C}^{2\times d_{\text{ph}}}$ with $C_{s,n_{\text{ph}}}$, take $U(k) = [\,u_\alpha(k)\,|\,u_\beta(k)\,]$ (columns = the SSH Bloch spinors of Step 7), and set

$$\boxed{\Psi_n(k) = U(k)\,C^{(n)}(k)\ \in\ \mathbb{C}^2_{\text{sub}} \otimes \mathbb{C}^{d_{\text{ph}}}_{\text{cav}}}$$

flattened sublattice-outer. Then $\langle\psi_n(k)|\psi_n(k')\rangle = \Psi_n(k)^\dagger \Psi_n(k')$ — an ordinary inner product. The BZ-boundary factor acts on the **sublattice half only** (the cavity does not wind):

$$\hat{g} = \text{diag}\left(e^{-2\pi i s_A}, e^{-2\pi i s_B}\right) \otimes 1_{d_{\text{ph}}} \quad\Longrightarrow\quad \hat{g}_{\text{flat}} = \texttt{repeat}\big([e^{-2\pi i s_A}, e^{-2\pi i s_B}],\, d_{\text{ph}}\big),$$

$$\gamma_{\text{Zak}}^{(n)} = -\,\text{Im}\ln\Big[\prod_{j=0}^{N_k-2} \Psi_n(k_j)^\dagger \Psi_n(k_{j+1})\ \cdot\ \Psi_n(k_{N_k-1})^\dagger\big(\hat{g}_{\text{flat}} \odot \Psi_n(k_0)\big)\Big].$$

Multiband version (for a near-degenerate set $\mathfrak{B}$, more robust than any single band): build the link matrices $\mathcal{M}^{(j)}_{nn'} = \Psi_n(k_j)^\dagger \Psi_{n'}(k_{j+1})$, take $W = \prod_j \mathcal{M}^{(j)}$ (with $\hat{g}$ on the last link) and $\gamma = -\,\text{Im}\ln\det W$.

Which band. Sort eigenvalues ascending. Band 0 is the zero-photon lower state $|\beta_k, 0\rangle$; the **second band** is index 1, the lower polariton of the anticrossing.

Step 12 — Cavity purity (diagnostic colouring)

Trace the band degree of freedom out of the eigenstate. With $C \in \mathbb{C}^{d_{\text{ph}}\times 2}$ indexed $[\,n_{\text{ph}},\ s\,]$,

$$\rho_{\text{ph}} = C\,C^\dagger\ (d_{\text{ph}} \times d_{\text{ph}}), \qquad \text{purity} = \text{Tr}\,\rho_{\text{ph}}^2 = \sum_{m,m'} \left|(\rho_{\text{ph}})_{mm'}\right|^2.$$

Range $[\frac{1}{2}, 1]$: 1 = pure Fock (band and cavity unentangled, a clean photon replica), $\frac{1}{2}$ = maximally band–photon hybridised (right at an anticrossing). Plot $\log_{10}(1 - \text{Tr}\,\rho_{\text{ph}}^2)$ (clipped at $\sim 10^{-6}$) to resolve the nearly-pure regions.

Results

Reference parameters

$N = 10$, $\Omega_C = 0.2$, $\chi_C = 0.5$, $\Omega_J = 1.0$, $\chi_J = 0 \Rightarrow g_{c1} = 0.075,\ g_{c2} = 0.025,\ g_{J1} = 1,\ g_{J2} = -1,\ g_J = 0.$ $\omega_r = 2(g_{c1} + g_{c2}) = 0.2$, $\gamma = 0.03$, $d_{\text{ph}} = 4$–6, $s = (-\frac{1}{4}, \frac{1}{4})$. Interband gap window $[\text{gap}(\pi), \text{gap}(0)] = [0.100,\ 0.201]$; $k^\star = 0.2315$.

Zak phases

object	γ_{Zak}
bare SSH lower band, $\chi_C > 0$ / $\chi_C < 0$	0 / π
BdG β quasiparticle band (pairing included)	identical, to 10^{-12}, in every gauge
dressed band 0 — $	\beta_k,0\rangle$
dressed band 1 — second band (lower polariton)	$\boldsymbol{\pi}$
dressed band 2 — upper polariton	π
dressed bands $\{1, 2\}$ combined (det Wilson loop)	0

- $\gamma = 0 \Rightarrow$ band 1 is trivial; **any** $\gamma > 0 \Rightarrow \pi$. The cavity gaps the $|\alpha_k, 0\rangle$–$|\beta_k, 1\rangle$ crossing and the second band becomes topological — the photon-induced edge state.
- The invariant tracks the SSH phase: $\chi_C > 0$ gives (band 0 trivial, band 1 topological), $\chi_C < 0$ the reverse. The dressing hands the invariant from band 0 to band 1.

Cavity frequency as a topological knob

$$\gamma_{\mathrm{Zak}}(\text{band } 1) = \pi \ \text{ for } \omega_r \lesssim \mathrm{gap}(0), \qquad = 0 \ \text{ for } \omega_r > \mathrm{gap}(0),$$

a clean step at $\omega_r \approx 0.207$. Above it no k satisfies the resonance condition, band 1 becomes pure $|\alpha_k, 0\rangle$, and the invariant reverts to the trivial bare-SSH value. The combined $\{1, 2\}$ invariant stays 0 throughout — the π is genuinely split between the two polaritons. Robust across $d_{\mathrm{ph}} = 3$–6 and N_k.

Photon replicas and the mechanism

The uncoupled ladder is the two SSH bands replicated at $\omega_{k\alpha/\beta} + m\omega_r$; the dressed bands follow them except at replica crossings, where they anticross. Colouring by cavity purity shows the mechanism directly:

- band 0 stays essentially pure across the whole BZ;
- the second band runs pure $|\beta_k, 1\rangle$ near the zone centre $\rightarrow$ 50/50 hybridised at $k^\star \approx \pm 2$ (exactly at the β/α replica crossing) $\rightarrow$ pure $|\alpha_k, 0\rangle$ near $k = \pi$. **That character swap across the BZ is the π Zak phase.**
- sharp purity spikes sit at $k = 0, \pm\pi$ — at the TRIMs the interband coupling $M_{\alpha\beta} \rightarrow 0$ and the bands cannot hybridise at all.

Caveats

1. **The SW correction is not a light dressing.** With $\Omega_J = 1$ the pairing block $\tilde{\mathcal{B}}_J$ is $O(1)$, so at $\gamma = 0.03$ one gets $\|H_{\mathrm{SW}}\| \sim 0.1$–$0.5$, comparable to the bandwidth, and it grows with d_{ph} because the $\Delta n_{\mathrm{ph}} = \pm 2$ terms carry $\sqrt{n}$. Conclusions were checked stable in d_{ph} and N_k, but the genuinely perturbative regime is $\gamma \sim 0.01$, where the with/without-SW spectra are nearly identical.
2. **The band 1 / band 2 gap at $k^\star$ is small** (10^{-3}–10^{-2}, set by $\gamma|M_{\alpha\beta}(k^\star)|$, and $k^\star$ sits near a TRIM where $M_{\alpha\beta}$ is itself small). The single-band invariant is therefore more delicate than the combined $\{1, 2\}$ one. Hardening it means widening the gap: larger γ, or an ω_r that puts $k^\star$ further from the TRIM.
3. **Not done yet:** an open-boundary real-space calculation to exhibit the predicted edge state and confirm bulk–boundary correspondence for the dressed second band.

4. **Open (being explored):** why H_{SW} has *exactly* zero $+k \leftrightarrow -k$ weight. Momentum conservation explains that a single quasiparticle cannot change k, but it is worth checking directly whether that is the full mechanism, whether the same holds for the number-conserving coupling and at higher order, and what it implies for the block structure of the effective theory.

Map to the notebook

object	function
Step 1 real-space Nambu blocks	`real_space_nambu`
Step 2 FFT momenta, $\pm k$ partner	`k_grid`, `neg_k_index`
Step 2 operator Fourier matrix $\mathcal{F}$, $\mathcal{G}^\dagger \mathcal{H} \mathcal{G}$	`fourier_operator_matrix`, `to_k_space_nambu`
Step 3 permutation + verification	`sector_permutation`, `offblock_norm`
Step 4 closed-form per-k blocks	`k_space_hamiltonians`, `reduced_k_blocks`, `sector_nambu_at_k`
Steps 5–6 rotation, squeeze, paraunitarity	`phased_sublattice_rotation`, `squeeze_nambu`, `passive_nambu`, `diagonalize_C_sector`, `para_defect`, `bogoliubov_freqs`
Step 7 Wilson loop	`compute_zak_phase`, `ssh_lower_vec`
Step 8 BdG spinor	`H_bdg_bloch`, `beta_bogoliubov_spinor`
Step 9 SW, composite	`H_SW_composite`
Step 9 SW, cavity-eliminated variant ($\delta A, \delta B$)	`sw_correction`
Step 9d parameter sweep	`peak_kchg`
Steps 10–11 dressed bands + composite Zak	`ssh_bloch_U`, `H_eff_dressed`, `H_eff_noSW`, `dressed_state`, `zak_dressed`
Step 12 purity colouring	`spec_purity`